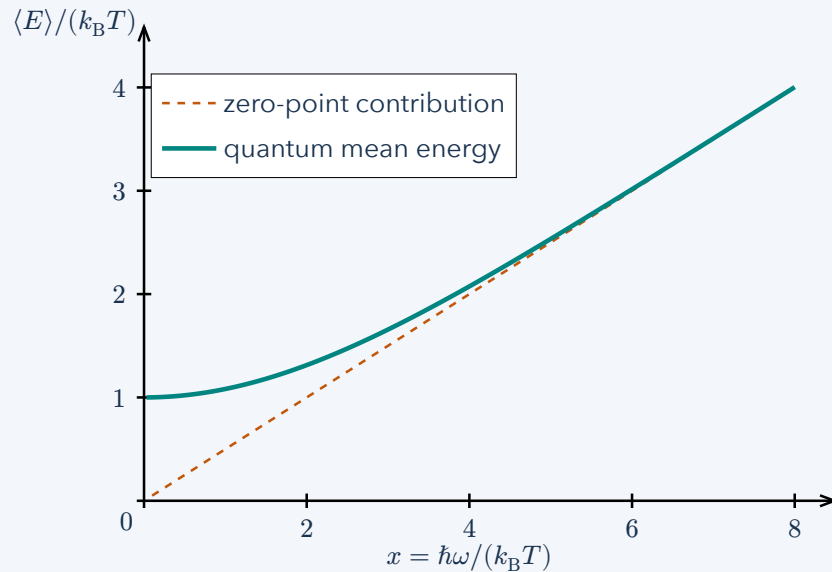


# Quantum Harmonic Oscillator

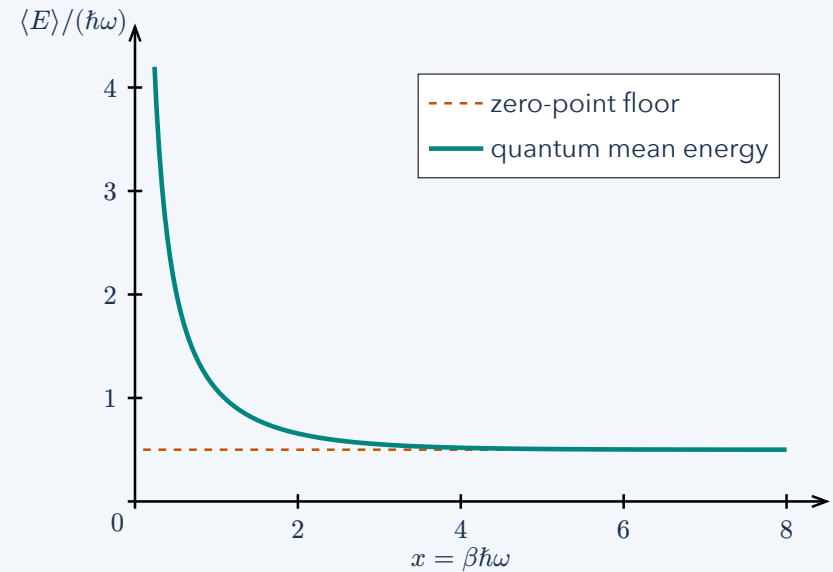
One energy formula connects thermal motion with zero-point motion. Compare the level spacing  $\hbar\omega$  with the thermal energy  $k_B T$ .

## 1 Increase frequency at fixed temperature



At small level spacing,  $\langle E \rangle \approx k_B T$ : classical equipartition. At large spacing the zero-point term dominates. The energy rises monotonically.

## 2 Cool at fixed frequency



Move right by increasing  $\beta = \frac{1}{k_B T}$ . Thermal excitations freeze out; the mean energy approaches  $\hbar\frac{\omega}{2}$ , rather than zero.

$\langle E \rangle = \hbar\omega \left( \frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1} \right)$ . **The same horizontal variable, two energy units.**  $\omega$  is angular frequency;  $\hbar$  is the reduced Planck constant.